

## Transparent radiative cooling film for cars and cabins

Researchers in South Korea have developed a transparent cooling film that can lower temperatures inside vehicles and electronic systems without consuming power, potentially improving energy efficiency in electric cars. The film



Conceptual illustration of vehicle application of large-area transparent radiative cooling film (Credit: Energy & Environmental Science)

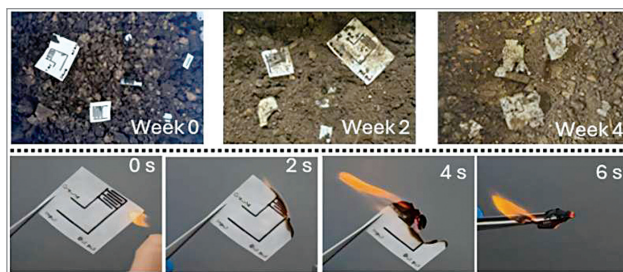
blocks heat-generating infrared radiation while maintaining high visible-light transparency, allowing clear visibility through windows and display covers. The passive radiative-cooling material dissipates heat into the atmosphere, reducing thermal buildup in automotive electronics and cabins. The technology could cut air-conditioning loads, extend battery range, and improve the reliability of sensitive electronics in hot conditions. The research also highlights scalability for flexible and automotive applications.

## Quantum chip converts current to sound for better computing

Researchers have developed a quantum chip that converts electrical current into sound waves, enabling quantum information to be processed via acoustic signals rather than conventional electronic methods. Engineered by a team at Yale University, the chip integrates superconducting qubits with acoustic resonators on a single platform. The device uses phonons, quantised sound vibrations, to transfer and store fragile quantum data with lower energy loss, potentially improving quantum computing stability and efficiency. The advance could support scalable quantum processors, secure communications, and next-generation sensing systems by improving the control and preservation of quantum states.

## Laser-based circuits on paper for disposable electronics

Researchers at Binghamton University have developed a laser-based method to create electronic circuits directly on parchment paper, enabling low-cost and biodegradable 'papertronics' for disposable electronics. The process uses a standard carbon dioxide laser to remove silicone coatings from paper, exposing cellulose fibres that absorb conductive inks. The technique forms resistors, capacitors, and interconnects without complex fabrication steps or silicon

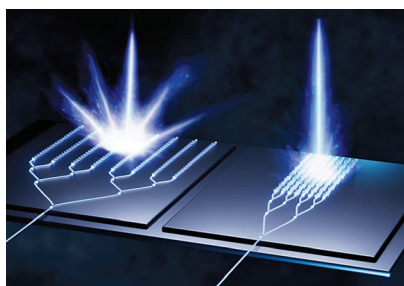


Paper-based electronics or papertronics can be composted or burned for disposal (Credit: [www.binghamton.edu](http://www.binghamton.edu))

substrates. The approach could accelerate the production of smart packaging, diagnostic strips, and environmental sensors while reducing electronic waste through recyclable, compostable paper-based electronics.

## Silicon photonics shrinks lidar hardware

Engineered by a team at the Massachusetts Institute of Technology, the chip integrates optical functions onto



Silicon Photonics shrinks lidar hardware (Credit: <https://news.mit.edu>)

semiconductor hardware, making lidar systems more compact and durable. Researchers have developed a silicon-photonics lidar chip that reduces hardware size while improving scanning range and accuracy

for autonomous systems. The design suppresses signal interference between optical antennas, enabling wider field-of-view scanning without bulky mechanical parts. The advance could lower costs and accelerate adoption in autonomous vehicles, robotics, and industrial sensing, where traditional lidar units remain expensive, large, and mechanically complex.

## Low-cost sensor helps see what eyes can't

Scientists from Daegu Gyeongbuk Institute of Science and Technology and partner institutes developed a new infrared sensor that detects weak infrared light. Traditional infrared cameras are expensive because they use costly materials, but this device uses different materials that work together to detect light and produce an electrical signal. The sensor was tested in a small system that can form images. It can be made using common manufacturing methods, enabling large-scale production at lower cost. This technology can be used in self-driving cars, security cameras, robots, and medical devices, especially in conditions like darkness, fog, or smoke.